

9.2. The General Lift Distribution

Prandtl's lifting-line equation can be solved, for a single finite wing with no sweep and no dihedral, by making use of a Fourier series expansion. To facilitate the Fourier expansion, we use the change of variables

$$z = -\frac{1}{2}b \cos(\theta) \quad (9.2.1)$$

which gives

$$dz = \frac{1}{2}b \sin(\theta)d\theta \quad (9.2.2)$$

With this change of variables, the spanwise coordinate, θ , varies along the wing in the range, $0 \leq \theta \leq \pi$. Since the circulation must be zero at both wingtips, this change of variables makes it convenient to use a Fourier sine series expansion for the circulation distribution along the span of the wing,

$$\Gamma(\theta) = 2bV_\infty \sum_{n=1}^{\infty} B_n \sin(n\theta) \quad (9.2.3)$$

Differentiating Eq. (9.2.3) with respect to z , we obtain

$$\frac{d\Gamma}{dz} = \frac{d\Gamma}{d\theta} \frac{d\theta}{dz} = 2bV_\infty \sum_{n=1}^{\infty} nB_n \cos(n\theta) \frac{d\theta}{dz} \quad (9.2.4)$$

Using Eqs. (9.2.1) through (9.2.4) in Eq. (9.1.24), we have

$$\frac{4b}{c(\theta)} \sum_{n=1}^{\infty} B_n \sin(n\theta) + \frac{\tilde{C}_{L,\alpha}}{\pi} \int_{\phi=0}^{\pi} \sum_{n=1}^{\infty} nB_n \frac{\cos(n\phi)d\phi}{\cos(\phi) - \cos(\theta)} = \tilde{C}_{L,\alpha}[\alpha(\theta) - \alpha_{L0}(\theta)] \quad (9.2.5)$$

where ϕ is the value of θ at $z = \zeta$. The integral in Eq. (9.2.5) can be evaluated to give,

$$\int_{\phi=0}^{\pi} \frac{\cos(n\phi)d\phi}{\cos(\phi) - \cos(\theta)} = \frac{\pi \sin(n\theta)}{\sin(\theta)} \quad (9.2.6)$$

Using Eq. (9.2.6) in Eq. (9.2.5), we have

$$\frac{4b}{\tilde{C}_{L,\alpha} c(\theta)} \sum_{n=1}^{\infty} B_n \sin(n\theta) + \sum_{n=1}^{\infty} nB_n \frac{\sin(n\theta)}{\sin(\theta)} = \alpha(\theta) - \alpha_{L0}(\theta) \quad (9.2.7)$$

or

$$\sum_{n=1}^{\infty} B_n \sin(n\theta) \left[\frac{4b}{\tilde{C}_{L,\alpha} c(\theta)} + \frac{n}{\sin(\theta)} \right] = \alpha(\theta) - \alpha_{L0}(\theta) \quad (9.2.8)$$

Eq. (9.2.8) is not in the form of a conventional Fourier series, so we cannot evaluate the coefficients using Fourier integrals. However, we can evaluate these coefficients numerically.

To evaluate the coefficients in Eq. (9.2.8) numerically, we must truncate the series at a finite number of terms,

$$\sum_{n=1}^N B_n \sin(n\theta) \left[\frac{4b}{\tilde{C}_{L,\alpha} c(\theta)} + \frac{n}{\sin(\theta)} \right] = \alpha(\theta) - \alpha_{L0}(\theta) \quad (9.2.9)$$

To solve for the N unknown coefficients, we force Eq. (9.2.9) to be satisfied at N specific sections along the wing. This gives us N equations in the N unknown coefficients B_n , which can be solved numerically.

Once the coefficients, B_n , in Eq. (9.2.3) have been determined, we can compute the total lift coefficient by using Eqs. (9.2.2) and (9.2.3) in Eq. (9.1.27)

$$C_L = \frac{2}{V_\infty S_p} \int_{z=-b/2}^{b/2} \Gamma(z) dz = \frac{2b^2}{S_p} \sum_{n=1}^N B_n \int_{\theta=0}^{\pi} \sin(n\theta) \sin(\theta) d\theta \quad (9.2.10)$$

The integral in Eq. (9.2.10) can be evaluated to give

$$\int_{\theta=0}^{\pi} \sin(n\theta) \sin(\theta) d\theta = \begin{cases} \pi/2, & n=1 \\ 0, & n \neq 1 \end{cases} \quad (9.2.11)$$

Using Eq. (9.2.11) in Eq. (9.2.10), we obtain

$$C_L = \frac{\pi b^2}{S_p} B_1 \quad (9.2.12)$$

Note that, even though the lift coefficient depends only on the first coefficient in the Fourier series, we must solve for all of the coefficients simultaneously in order to obtain the first coefficient, B_1 . Equation (9.2.12) introduces an important geometric property of a finite wing, that is the aspect ratio, denoted by R_A and defined as

$$R_A \equiv \frac{b^2}{S_p} \quad (9.2.13)$$

Hence, Eq. (9.2.12) becomes

$$C_L = \pi R_A B_1 \quad (9.2.14)$$

The induced drag coefficient is obtained by using Eqs. (9.2.1) through (9.2.4) in Eq. (9.1.31). This gives

$$\begin{aligned} C_{D_i} &= \int_{z=-b/2}^{b/2} \frac{\Gamma(z)}{2\pi V_\infty^2 S_p} \int_{\zeta=-b/2}^{b/2} \frac{d\Gamma}{d\zeta} \frac{d\zeta}{z-\zeta} dz \\ &= \frac{2b^2}{\pi S_p} \int_{\theta=0}^{\pi} \sum_{n=1}^N B_n \sin(n\theta) \sum_{n=1}^N n B_n \int_{\phi=0}^{\pi} \frac{\cos(n\phi) d\phi}{\cos(\phi) - \cos(\theta)} \sin(\theta) d\theta \end{aligned} \quad (9.2.15)$$

The integration with respect to ϕ can be expressed using Eq. (9.2.6). Thus, Eq. (9.2.15) can be written

$$C_{D_i} = \frac{2b^2}{S_p} \int_{\theta=0}^{\pi} \left[\sum_{n=1}^N B_n \sin(n\theta) \right] \left[\sum_{n=1}^N n B_n \sin(n\theta) \right] d\theta \quad (9.2.16)$$

Equation (9.2.16) involves the product of two summations. However, recall that

$$\int_{\theta=0}^{\pi} \sin(n\theta) \sin(m\theta) d\theta = \begin{cases} \pi/2, & n=m \\ 0, & n \neq m \end{cases} \quad (9.2.17)$$

which means that, all of the mixed product terms involving unequal subscripts are zero. Thus, using Eq. (9.2.17), we can write Eq. (9.2.16) as

$$C_{Di} = \frac{\pi b^2}{S_p} \sum_{n=1}^N n B_n^2 = \pi R_A B_1^2 \left[1 + \sum_{n=2}^N n \left(\frac{B_n}{B_1} \right)^2 \right] \quad (9.2.18)$$

Using Eq. (9.2.14), in Eq. (9.2.18) gives

$$C_{Di} = \frac{C_L^2}{\pi R_A} \left[1 + \sum_{n=2}^N n \left(\frac{B_n}{B_1} \right)^2 \right] = \frac{C_L^2}{\pi R_A} (1 + \kappa_D) = \frac{C_L^2}{\pi R_A e_s} \quad (9.2.19)$$

where

$$\kappa_D \equiv \sum_{n=2}^N n \left(\frac{B_n}{B_1} \right)^2 \quad (9.2.20)$$

and

$$e_s \equiv \frac{1}{1 + \kappa_D} \quad (9.2.21)$$

Note that κ_D is always greater than or equal to zero and e_s is always less than or equal to unity, which means that the minimum induced drag occurs for a wing with $e_s = 1$. Thus, the parameter e_s is called the span efficiency factor.

From Eq. (9.2.19) we see that the induced drag coefficient is directly proportional to the square of the lift coefficient and inversely proportional to the aspect ratio of the wing. This is a very important aerodynamic relationship that will be used extensively in upcoming chapters of this text. Ludwig Prandtl's lifting-line theory was the first theory of finite wings to predict this classic relation. Prandtl (1921) was also the first to verify this relation with a classic set of aerodynamic experiments. In 1915, Prandtl measured the lift and drag on seven rectangular wings of different aspect ratio, varying from 1.0 to 7.0, and plotted the drag coefficient as a function of lift coefficient for each wing. These experimental results confirmed the parabolic variation in drag coefficient with lift coefficient that is predicted by Eq. (9.2.19). The inverse relationship between induced drag and aspect ratio was also confirmed.

The dependence of drag on lift should not be surprising. The drag is the component of the aerodynamic force that must be countered by the thrust of the airplane's engine. Thus, we should not expect the drag to be independent of the lift. Rather, we should expect the generation of increased lift to produce increased drag and thus require increased thrust. The induced drag is that portion of the total drag that is a direct result of the lift produced by the wing and is sometimes called the drag due to lift. Since the induced drag coefficient is proportional to the square of the lift coefficient, the induced drag increases rapidly as the lift coefficient increases. For aircraft flying slowly in a high-lift configuration, such as during takeoff or landing, the induced drag can be the major component of the total drag. Even for an aircraft flying in cruise configuration, at or near the minimum drag airspeed, the induced drag is approximately one-half of the total drag.

The inverse relationship between induced drag and aspect ratio has a profound effect on aircraft design. To decrease the induced drag as much as possible, a subsonic aircraft needs to have a wing with the highest possible aspect ratio. However, the design of high aspect ratio wings with sufficient strength and rigidity to support the weight of a large aircraft is extremely difficult. Thus, the aspect ratio of a conventional airplane is usually a compromise between aerodynamic efficiency and structural considerations. The aspect ratio of most conventional airplanes is in the range from about 5 to 10, with some specialty aircraft like sailplanes having aspect ratios as high as 25.

Another important consideration that can be obtained from finite wing theory is the downwash induced by the wing on other surfaces of the aircraft. This downwash significantly affects the aerodynamic forces and moments on the aircraft and thus must be considered in aircraft design. A typical downwash distribution for flow around the wing of a conventional airplane is shown schematically in Fig. 9.2.1.

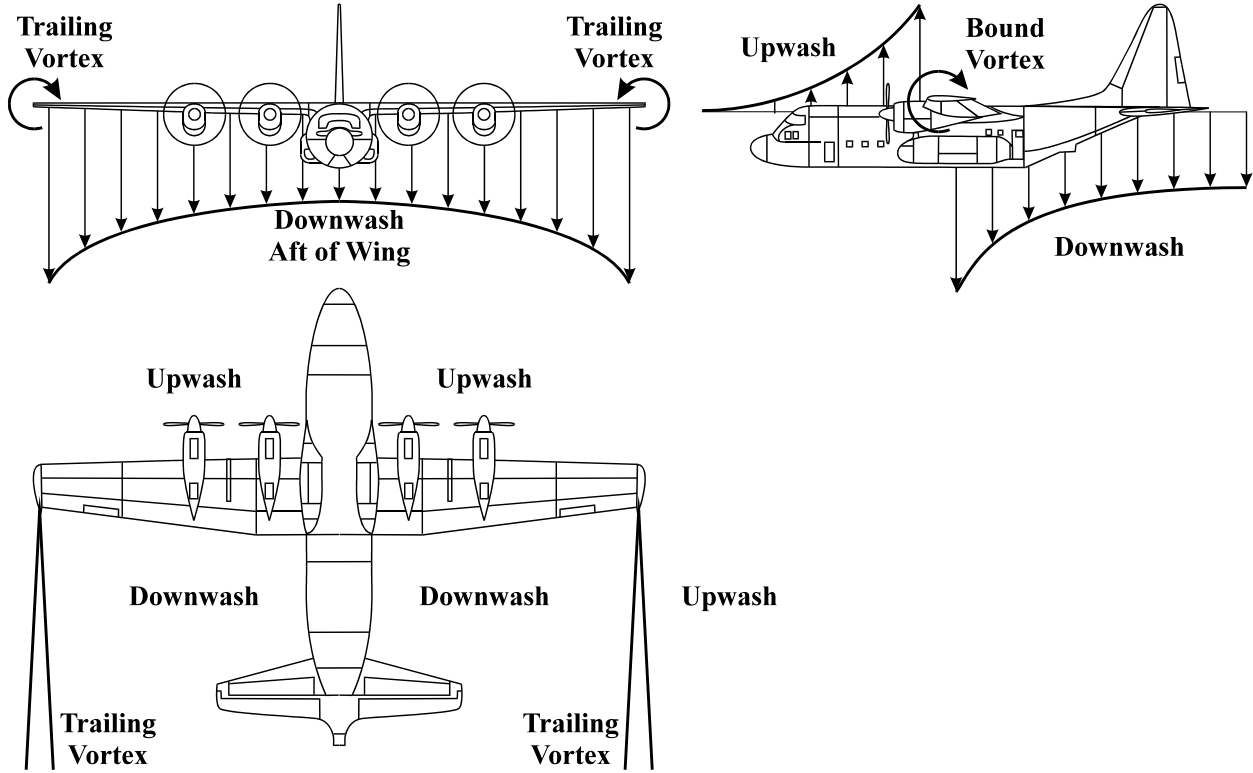


Figure 9.2.1. The downwash distribution created by the wing of a conventional airplane.

The downwash induced by a finite wing at any point near the wing is found by using Eqs. (9.2.1) through (9.2.4) in Eq. (9.1.17). If we also introduce the dimensionless coordinates,

$$\bar{x} = \frac{x}{b/2} \quad (9.2.22)$$

$$\bar{y} = \frac{y}{b/2} \quad (9.2.23)$$

$$\bar{z} = \frac{z}{b/2} \quad (9.2.24)$$

we have,

$$V_y(\bar{x}, \bar{y}, \bar{z}) = -\frac{V_\infty}{\pi} \sum_{n=1}^N B_n \int_{\phi=0}^{\pi} \left\{ \frac{n \cos(n\phi) [\bar{z} + \cos(\phi)]}{\bar{y}^2 + [\bar{z} + \cos(\phi)]^2} + \frac{n \bar{x} \cos(n\phi) [\bar{z} + \cos(\phi)]}{\{\bar{y}^2 + [\bar{z} + \cos(\phi)]^2\} \{\bar{x}^2 + \bar{y}^2 + [\bar{z} + \cos(\phi)]^2\}^{1/2}} + \frac{\bar{x} \sin(n\phi) \sin(\phi)}{\{\bar{x}^2 + \bar{y}^2 + [\bar{z} + \cos(\phi)]^2\}^{3/2}} \right\} d\phi \quad (9.2.25)$$

or in view of Eq. (9.2.14),

$$V_y(\bar{x}, \bar{y}, \bar{z}) = -\frac{V_\infty C_L}{\pi R_A} \sum_{n=1}^N \frac{B_n}{B_1} \frac{1}{\pi} \int_{\phi=0}^{\pi} \left\{ \frac{n \cos(n\phi) [\bar{z} + \cos(\phi)]}{\bar{y}^2 + [\bar{z} + \cos(\phi)]^2} + \frac{n \bar{x} \cos(n\phi) [\bar{z} + \cos(\phi)]}{\{\bar{y}^2 + [\bar{z} + \cos(\phi)]^2\} \{\bar{x}^2 + \bar{y}^2 + [\bar{z} + \cos(\phi)]^2\}^{1/2}} + \frac{\bar{x} \sin(n\phi) \sin(\phi)}{\{\bar{x}^2 + \bar{y}^2 + [\bar{z} + \cos(\phi)]^2\}^{3/2}} \right\} d\phi \quad (9.2.26)$$

Two special cases of Eq. (9.2.26) are of particular interest, the downwash along the wing axis,

$$V_y(0, 0, \bar{z}) = -\frac{V_\infty C_L}{\pi R_A} \sum_{n=1}^N \frac{B_n}{B_1} \frac{1}{\pi} \int_{\phi=0}^{\pi} \frac{n \cos(n\phi)}{\bar{z} + \cos(\phi)} d\phi \quad (9.2.27)$$

and the downwash along the fuselage axis,

$$V_y(\bar{x}, 0, 0) = -\frac{V_\infty C_L}{\pi R_A} \sum_{n=1}^N \frac{B_n}{B_1} \frac{1}{\pi} \int_{\phi=0}^{\pi} \left\{ \frac{n \cos(n\phi)}{\cos(\phi)} + \frac{n \bar{x} \cos(n\phi)}{\cos(\phi) [\bar{x}^2 + \cos^2(\phi)]^{1/2}} + \frac{\bar{x} \sin(n\phi) \sin(\phi)}{[\bar{x}^2 + \cos^2(\phi)]^{3/2}} \right\} d\phi \quad (9.2.28)$$

Equation (9.2.26) gives a good representation of the downwash in the region close to a finite wing. As we move above, below, or in front of the wing, the downwash goes to zero as the distance from the wing increases. This is represented by Eq. (9.2.26). On the other hand, as we move behind the wing, Eq. (9.2.26) predicts that the downwash will remain finite and become independent of x . This is, at least qualitatively, observed in fact. The downwash associated with the vorticity generated by a finite wing remains significant for several miles behind a large aircraft. However, the downwash distribution predicted by Eq. (9.2.26) is not accurate in the region far behind a wing. This is because Eq. (9.2.26) is based on the assumption that the vortex filaments trailing downstream from the wing are all straight and parallel to the freestream flow, as was shown in Figs. 9.1.5 and 9.1.6. In reality, the vorticity trailing from each side of the wing will rollup around an axis trailing slightly inboard from the wingtip, as is shown schematically in Fig. 9.2.2.

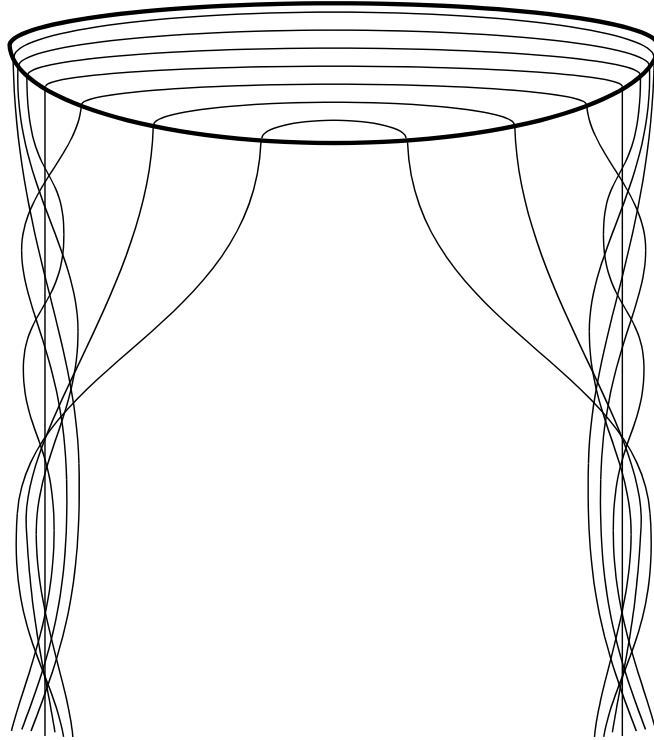


Figure 9.2.2. Schematic of the vorticity rollup behind a finite wing with elliptic planform shape.

The rollup of the free vortex sheet trailing behind each semi-span of the wing can be viewed as a result of the three-dimensional vortex lifting law developed in App. F. This vortex lifting law requires that, in any potential flow containing vortex filaments, a force is exerted on the surroundings that is proportional to the cross product of the local fluid velocity with the local filament vorticity. Since a free vortex filament cannot support a force, the cross product of the local fluid velocity with the local filament vorticity must be zero at every point along a free vortex filament. This means that all free vortex filaments must everywhere follow the streamlines of the flow. Thus, the free vortex filaments trailing behind each semi-span of the wing will rollup exactly as the streamlines were seen to rollup in Fig. 9.1.4.

Within a few chord lengths behind the wing, the vortex sheet becomes completely rolled-up to form two large counter-rotating vortices, one trailing behind each wing. This rollup has a significant effect on the aerodynamic forces and moments exerted on any surface of an aircraft that is behind the wing. Thus, it is important to be able to predict the strength and position of these wingtip vortices and to be able to predict the downwash that is induced on other surfaces of an aircraft by these vortices.

Each wingtip vortex is generated from the trailing vortex sheet produced by one half of the wing. Therefore, a wingtip vortex, a few chord lengths or more behind the wing, can be approximated as a single vortex of strength, Γ_{wt} , which is given by

$$\Gamma_{wt} = \int_{z=0}^{b/2} \gamma_t(z) dz \quad (9.2.29)$$

or in view of Eq. (9.1.2)

$$\Gamma_{wt} = - \int_{z=0}^{b/2} \frac{d\Gamma}{dz} dz = -[\Gamma(b/2) - \Gamma(0)] = \Gamma(0) \quad (9.2.30)$$

Using Eqs. (9.2.1) and (9.2.3) in Eq. (9.2.30), we have

$$\Gamma_{wt} = 2bV_\infty \sum_{n=1}^N B_n \sin(n\pi/2) \quad (9.2.31)$$

or in view of Eq. (9.2.14)

$$\Gamma_{wt} = \frac{2bV_\infty C_L}{\pi R_A} \left[1 + \sum_{n=2}^N \frac{B_n}{B_1} \sin(n\pi/2) \right] \quad (9.2.32)$$

When computing the downwash a few chord lengths or more downstream from a finite wing, we can approximate the vortex sheet as a single horseshoe shaped vortex filament of strength Γ_{wt} , as shown in Fig. 9.2.3. The distance between the trailing vortices, b' , is less than the wingspan because the vortex sheet from each side of the wing rolls up around the center of vorticity, which is slightly inboard from the wingtip. The horseshoe filament starts an infinite distance downstream from a point slightly inboard of the left wingtip, $(\infty, 0, b'/2)$, and runs upstream along the left wingtip vortex to the left wing, $(0, 0, b'/2)$. From there it runs across the wing quarter-chord to a point slightly inboard of the right wingtip, $(0, 0, -b'/2)$, and downstream along the right wingtip vortex to infinity, $(\infty, 0, -b'/2)$. The velocity induced by this vortex filament can be obtained from the Biot-Savart law.

Consider a differential segment of the left wingtip vortex located at the point $(\xi, 0, b'/2)$ and having a length $d\xi$. The directed segment length for this differential segment is

$$d\vec{l} = -d\xi \vec{i}_x + 0\vec{i}_y + 0\vec{i}_z \quad (9.2.33)$$

and the radius vector from the filament at $(\xi, 0, b'/2)$ to a point at (x, y, z) is

$$\vec{r} = (x - \xi)\vec{i}_x + y\vec{i}_y + (z - \frac{1}{2}b')\vec{i}_z \quad (9.2.34)$$

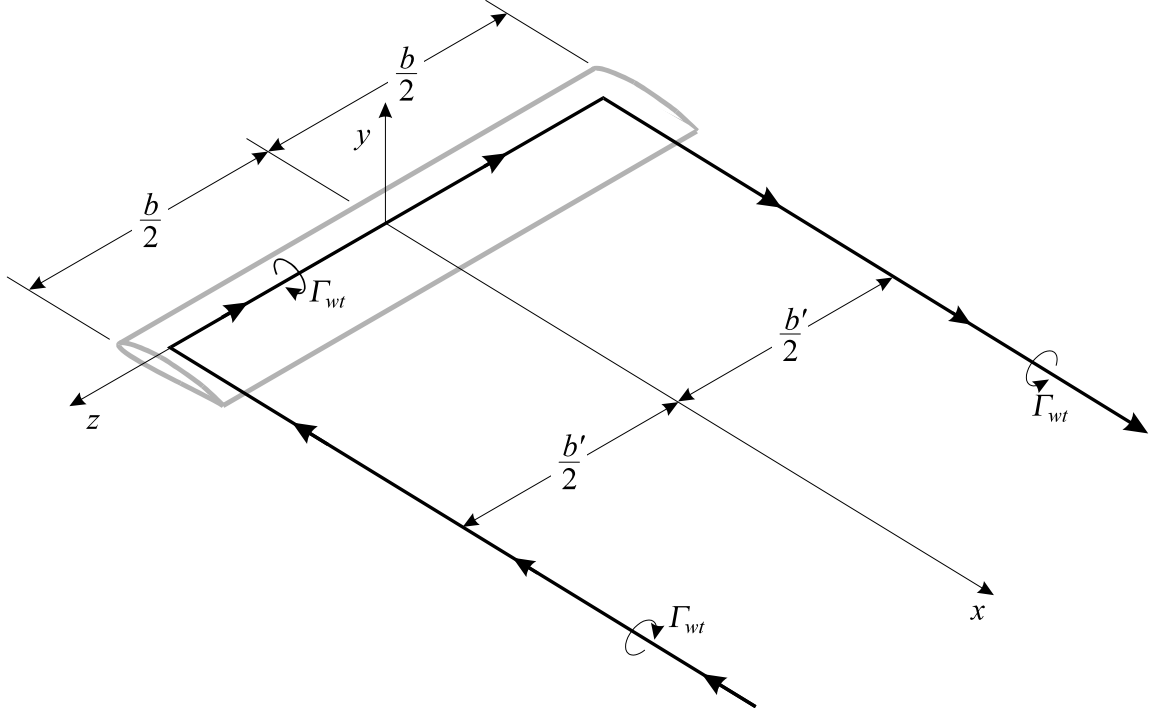


Figure 9.2.3. Vortex model for estimating the downwash a few chord lengths or more downstream.

Using Eqs. (9.2.33) and (9.2.34) in Eq. (9.1.1) and integrating, we obtain the total y -velocity component that is induced at the point (x, y, z) by the entire left wingtip vortex,

$$\begin{aligned}
 V_{Ly}(x, y, z) &= \frac{\Gamma_{wt}}{4\pi} \int_{\xi=0}^{\infty} \frac{(z - \frac{1}{2}b')d\xi}{[(x - \xi)^2 + y^2 + (z - \frac{1}{2}b')^2]^{3/2}} \\
 &= -\frac{\Gamma_{wt}}{4\pi} \frac{\frac{1}{2}b' - z}{y^2 + (\frac{1}{2}b' - z)^2} \left[1 + \frac{x}{\sqrt{x^2 + y^2 + (\frac{1}{2}b' - z)^2}} \right] \quad (9.2.35)
 \end{aligned}$$

For that portion of the vortex filament that is bound to the wing, the directed differential segment length is

$$d\vec{l} = 0\vec{i}_x + 0\vec{i}_y - d\zeta\vec{i}_z \quad (9.2.36)$$

the radius vector from the filament at $(0, 0, \zeta)$ to a point at (x, y, z) is

$$\vec{r} = x\vec{i}_x + y\vec{i}_y + (z - \zeta)\vec{i}_z \quad (9.2.37)$$

and the induced y -velocity component is

$$\begin{aligned}
 V_{By}(x, y, z) &= -\frac{\Gamma_{wt}}{4\pi} \int_{\zeta=-b'/2}^{b'/2} \frac{x d\zeta}{[x^2 + y^2 + (z - \zeta)^2]^{3/2}} \\
 &= -\frac{\Gamma_{wt}}{4\pi} \frac{x}{x^2 + y^2} \left[\frac{\frac{1}{2}b' - z}{\sqrt{x^2 + y^2 + (\frac{1}{2}b' - z)^2}} + \frac{\frac{1}{2}b' + z}{\sqrt{x^2 + y^2 + (\frac{1}{2}b' + z)^2}} \right] \quad (9.2.38)
 \end{aligned}$$

Similarly, for the right wingtip vortex, the directed differential segment length is

$$d\vec{l} = d\xi \vec{i}_x + 0\vec{i}_y + 0\vec{i}_z \quad (9.2.39)$$

the radius vector from the filament at $(\xi, 0, -b'/2)$ to a point at (x, y, z) is

$$\vec{r} = (x - \xi)\vec{i}_x + y\vec{i}_y + (z + \frac{1}{2}b')\vec{i}_z \quad (9.2.40)$$

and the induced y -velocity component is

$$\begin{aligned} V_{Ry}(x, y, z) &= -\frac{\Gamma_{wt}}{4\pi} \int_{\xi=0}^{\infty} \frac{(z + \frac{1}{2}b')d\xi}{[(x - \xi)^2 + y^2 + (z + \frac{1}{2}b')^2]^{3/2}} \\ &= -\frac{\Gamma_{wt}}{4\pi} \frac{\frac{1}{2}b' + z}{y^2 + (\frac{1}{2}b' + z)^2} \left[1 + \frac{x}{\sqrt{x^2 + y^2 + (\frac{1}{2}b' + z)^2}} \right] \end{aligned} \quad (9.2.41)$$

Combining Eqs. (9.2.35), (9.2.38) and (9.2.41), the total y -velocity component that is induced by the entire horseshoe vortex is

$$\begin{aligned} V_y(x, y, z) &= -\frac{\Gamma_{wt}}{4\pi} \left\{ \frac{\frac{1}{2}b' - z}{y^2 + (\frac{1}{2}b' - z)^2} \left[1 + \frac{x}{\sqrt{x^2 + y^2 + (\frac{1}{2}b' - z)^2}} \right] \right. \\ &\quad + \frac{x}{x^2 + y^2} \left[\frac{\frac{1}{2}b' - z}{\sqrt{x^2 + y^2 + (\frac{1}{2}b' - z)^2}} + \frac{\frac{1}{2}b' + z}{\sqrt{x^2 + y^2 + (\frac{1}{2}b' + z)^2}} \right] \\ &\quad \left. + \frac{\frac{1}{2}b' + z}{y^2 + (\frac{1}{2}b' + z)^2} \left[1 + \frac{x}{\sqrt{x^2 + y^2 + (\frac{1}{2}b' + z)^2}} \right] \right\} \end{aligned} \quad (9.2.42)$$

or in terms of the dimensionless coordinates

$$\begin{aligned} V_y(\bar{x}, \bar{y}, \bar{z}) &= -\frac{\Gamma_{wt}}{2\pi b} \left\{ \frac{\kappa_b - \bar{z}}{\bar{y}^2 + (\kappa_b - \bar{z})^2} \left[1 + \frac{\bar{x}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b - \bar{z})^2}} \right] \right. \\ &\quad + \frac{\bar{x}}{\bar{x}^2 + \bar{y}^2} \left[\frac{\kappa_b - \bar{z}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b - \bar{z})^2}} + \frac{\kappa_b + \bar{z}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b + \bar{z})^2}} \right] \\ &\quad \left. + \frac{\kappa_b + \bar{z}}{\bar{y}^2 + (\kappa_b + \bar{z})^2} \left[1 + \frac{\bar{x}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b + \bar{z})^2}} \right] \right\} \end{aligned} \quad (9.2.43)$$

where the dimensionless parameter κ_b is defined to be the ratio of the wingtip vortex spacing, b' , to the span of the finite wing, b ,

$$\kappa_b \equiv \frac{b'}{b}$$

Using Eq. (9.2.32) in Eq. (9.2.43), we can express the local downwash a few chord lengths or more downstream from a finite wing in terms of the free stream velocity and the lift coefficient,

$$V_y(\bar{x}, \bar{y}, \bar{z}) = -\frac{V_\infty C_L \kappa_v}{\pi^2 R_A} \left\{ \frac{\kappa_b - \bar{z}}{\bar{y}^2 + (\kappa_b - \bar{z})^2} \left[1 + \frac{\bar{x}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b - \bar{z})^2}} \right] \right. \\ \left. + \frac{\bar{x}}{\bar{x}^2 + \bar{y}^2} \left[\frac{\kappa_b - \bar{z}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b - \bar{z})^2}} + \frac{\kappa_b + \bar{z}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b + \bar{z})^2}} \right] \right. \\ \left. + \frac{\kappa_b + \bar{z}}{\bar{y}^2 + (\kappa_b + \bar{z})^2} \left[1 + \frac{\bar{x}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b + \bar{z})^2}} \right] \right\} \quad (9.2.44)$$

where

$$\kappa_v = 1 + \sum_{n=2}^N \frac{B_n}{B_1} \sin(n\pi/2) \quad (9.2.45)$$

Since the vortex sheet trailing behind each semi-span of the wing rolls up about the center of vorticity, we have

$$\frac{1}{2} b' = \int_{z=0}^{b/2} z \gamma_t(z) dz \bigg/ \int_{z=0}^{b/2} \gamma_t(z) dz \quad (9.2.46)$$

Using Eqs. (9.1.2) and (9.2.29) in Eq. (9.2.46) gives

$$\frac{1}{2} b' = \frac{-1}{\Gamma_{wt}} \int_{z=0}^{b/2} z \frac{d\Gamma}{dz} dz \quad (9.2.47)$$

Now, applying Eqs. (9.2.1), (9.2.4), and (9.2.31), this can be rewritten as

$$\frac{1}{2} b' = \frac{1}{2} b \frac{\sum_{n=1}^N n B_n \int_{\theta=\pi/2}^{\pi} \cos(n\theta) \cos(\theta) d\theta}{\sum_{n=1}^N B_n \sin(n\pi/2)} \quad (9.2.48)$$

The integration with respect to θ in Eq. (9.2.48) is readily carried out to give

$$\int_{\theta=\pi/2}^{\pi} \cos(n\theta) \cos(\theta) d\theta = \begin{cases} \frac{\pi}{4}, & n = 1 \\ \frac{\cos(n\pi/2)}{(n^2 - 1)}, & n \neq 1 \end{cases} \quad (9.2.49)$$

Using Eq. (9.2.49) in Eq. (9.2.48) results in

$$\kappa_b = \frac{b'}{b} = \frac{\frac{\pi}{4} + \sum_{n=2}^N \frac{n B_n}{(n^2 - 1) B_1} \cos(n\pi/2)}{1 + \sum_{n=2}^N \frac{B_n}{B_1} \sin(n\pi/2)} = \frac{1}{\kappa_v} \left[\frac{\pi}{4} + \sum_{n=2}^N \frac{n B_n}{(n^2 - 1) B_1} \cos(n\pi/2) \right] \quad (9.2.50)$$

In **summary**, an analytical solution to Prandtl's lifting-line equation can be obtained in terms of a Fourier sine series, for a single finite wing with no sweep and no dihedral. For a given wing design, at a given angle of attack, the planform shape, $c(z)$, the geometric twist $\alpha(z)$, and the aerodynamic twist, $\alpha_{L0}(z)$ are all known as functions of spanwise position. The circulation distribution can be written as a Fourier series, where the Fourier coefficients, B_n , are determined by forcing the lifting-line equation to be satisfied at N specific sections along the span of the wing.

The **circulation distribution** is given by

$$\Gamma(\theta) = 2bV_\infty \sum_{n=1}^N B_n \sin(n\theta); \quad \theta = \cos^{-1}(-2z/b) \quad (9.2.3)$$

$$\sum_{n=1}^N B_n \sin(n\theta) \left[\frac{4b}{\tilde{C}_{L,\alpha} c(\theta)} + \frac{n}{\sin(\theta)} \right] = \alpha(\theta) - \alpha_{L0}(\theta) \quad (9.2.9)$$

The **lift coefficient** for a finite wing can be expressed as

$$C_L = \pi R_A B_1; \quad R_A = \frac{b^2}{S_p} \quad (9.2.14)$$

The **induced drag coefficient** for a finite wing is

$$C_{Di} = \frac{C_L^2}{\pi R_A} (1 + \kappa_D) = \frac{C_L^2}{\pi R_A e_s}, \quad \kappa_D = \sum_{n=2}^N n \left(\frac{B_n}{B_1} \right)^2, \quad e_s = \frac{1}{1 + \kappa_D} \quad (9.2.19)$$

The **downwash induced near the wing** is given by

$$\begin{aligned} V_y(\bar{x}, \bar{y}, \bar{z}) = & -\frac{V_\infty C_L}{\pi R_A} \sum_{n=1}^N \frac{B_n}{B_1} \frac{1}{\pi} \int_{\phi=0}^{\pi} \left\{ \frac{n \cos(n\phi) [\bar{z} + \cos(\phi)]}{\bar{y}^2 + [\bar{z} + \cos(\phi)]^2} \right. \\ & \left. + \frac{n \bar{x} \cos(n\phi) [\bar{z} + \cos(\phi)]}{\{\bar{y}^2 + [\bar{z} + \cos(\phi)]^2\} \{\bar{x}^2 + \bar{y}^2 + [\bar{z} + \cos(\phi)]^2\}^{1/2}} + \frac{\bar{x} \sin(n\phi) \sin(\phi)}{\{\bar{x}^2 + \bar{y}^2 + [\bar{z} + \cos(\phi)]^2\}^{3/2}} \right\} d\phi \quad (9.2.26) \\ \bar{x} = & \frac{x}{b/2}, \quad \bar{y} = \frac{y}{b/2}, \quad \bar{z} = \frac{z}{b/2} \end{aligned}$$

and the **downwash induced a few chord lengths or more downstream** from the wing is

$$\begin{aligned} V_y(\bar{x}, \bar{y}, \bar{z}) = & -\frac{V_\infty C_L \kappa_v}{\pi^2 R_A} \left\{ \frac{\kappa_b - \bar{z}}{\bar{y}^2 + (\kappa_b - \bar{z})^2} \left[1 + \frac{\bar{x}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b - \bar{z})^2}} \right] \right. \\ & \left. + \frac{\bar{x}}{\bar{x}^2 + \bar{y}^2} \left[\frac{\kappa_b - \bar{z}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b - \bar{z})^2}} + \frac{\kappa_b + \bar{z}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b + \bar{z})^2}} \right] + \frac{\kappa_b + \bar{z}}{\bar{y}^2 + (\kappa_b + \bar{z})^2} \left[1 + \frac{\bar{x}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\kappa_b + \bar{z})^2}} \right] \right\} \quad (9.2.44) \\ \kappa_v = & 1 + \sum_{n=2}^N \frac{B_n}{B_1} \sin(n\pi/2), \quad \kappa_b = \frac{1}{\kappa_v} \left[\frac{\pi}{4} + \sum_{n=2}^N \frac{n B_n}{(n^2 - 1) B_1} \cos(n\pi/2) \right] \end{aligned}$$